

Project Proposal

Title:

- Active Recall Flashcard Reviewer

Problem Statement:

- Many students struggle to remember information efficiently because they rely on passive study methods instead of active recall. While digital flashcard applications exist to help, they are often bloated with unnecessary features and require constant internet access. There is a clear need for a highly accessible, simple offline tool that allows users to quickly create and manage custom study materials. This project is the embodiment of that by providing a lightweight, terminal-based Python program specifically designed for straightforward self-testing, memorization, and performance tracking.

Project Objectives:

Develop a straightforward study tool:

- Create a lightweight, terminal-based Python program that runs locally without complex installations.

Facilitate Active Recall:

- Prompt users to actively retrieve information from memory rather than just recognizing it.

Enable Customization:

- Allow users to create and manage multiple personalized study decks containing their own questions and answers.

Provide Immediate Feedback:

- Show correct answers immediately upon a wrong guess to reinforce learning.

Track Performance:

- Implement a scoring system that tracks correct answers and gives users a final score to measure their retention.

Planned Features:

Deck Management:

- The ability to create multiple different decks for different subjects and topics.

Flashcard Creation:

- A simple prompt system to input custom Questions and corresponding answers.

Playing the decks:

- A quizzing interface that presents questions one by one and waits for the user's answer.

Answer Validation & Feedback:

- Automatically checks the user's input against the stored answer, notifies them if they are right or wrong, and reveals the correct answer if they make a mistake.

Score Tracking:

- Totals the user's correct answers and displays a final score (Ex. Score: 4/5) at the end of the session.

Replayability:

- Provides a prompt to immediately replay the deck to reinforce missed questions.

Planned Inputs and Outputs:

Inputs:

- Navigation commands, which are used to control the program flow and navigate menus. (Ex. yes or no)
- Study data, which are the custom questions and correct answers entered by the user to build a deck. (Ex. Q: What's the capital of France?, A: Paris)
- The saved answer when creating a card for a deck presented to the user during the quizzing phase, for you to answer and input. (Ex. Beijing)

Outputs:

- System prompts, which are instructions guiding the user on what action is required next. (Ex. Make a new deck:)
- The saved question when creating a card for a deck presented to the user during the quizzing phase, for you to see and answer. (Ex. Q: What's the capital of France?)
- Feedback, which is the system's real-time evaluation of a guess, revealing the correct answer if a mistake is made. (Ex. Wrong Answer! Correct answer: Paris)
- The results, which is the final score from the deck played. (Ex. Final score: 0/1)

Logic Plan (Flowchart):

Flow
chart CS

